

Introduction

CodiceFiscale is a .NET 9 library for working with Italian personal identifiers.

What it does

Feature	Class	Status
Validate a Codice Fiscale	<code>CodiceFiscaleValidator</code>	✓ Available
Parse a Codice Fiscale into personal data	<code>CodiceFiscaleParser</code>	✓ Available
Generate a Codice Fiscale from personal data	<code>CodiceFiscaleGenerator</code>	➡ Coming soon
Match a Codice Fiscale against personal data	<code>CodiceFiscaleMatcher</code>	➡ Coming soon
Validate an IBAN	<code>IBANValidator</code>	✓ Available
Validate an Italian VAT code (Partita IVA)	<code>ItalianVatCodeValidator</code>	✓ Available

Design principles

- **Local-only** — all logic runs in-process. No HTTP calls, no external services. Personal data never leaves your application.
- **Embedded datasets** — the municipality/Belfiore dataset (~7 800 comuni) and the foreign country codes (~260 countries) are compiled directly into the DLL as embedded resources. No files to deploy, no database to configure.
- **Lazy loading** — datasets are deserialized once on first use and cached for the lifetime of the process. Repeated calls are fast.
- **Immutable results** — all output types are C# `records` with `init`-only properties.

Supported platforms

- .NET 9.0+

License

MIT — see the [repository](#) for details.

Getting Started

Installation

```
dotnet add package CodiceFiscale
```

Or via the NuGet Package Manager in Visual Studio: search for **CodiceFiscale**.

Validate a Codice Fiscale

```
using CodiceFiscale;

bool valid = CodiceFiscaleValidator.IsValid("RSSMRA85T10A562S");
// true

// Case-insensitive
bool valid2 = CodiceFiscaleValidator.IsValid("rSSmra85t10a562s");
// true

// Foreign birth place (Z-code)
bool validForeign = CodiceFiscaleValidator.IsValid("MRNGRL01P55Z614X");
// true

// Invalid check digit
bool invalid = CodiceFiscaleValidator.IsValid("RSSMRA85T10A562X");
// false
```

`IsValid` returns `false` (never throws) for any invalid input including `null`-equivalent strings, wrong length, bad characters, invalid date, unknown Belfiore code, or wrong check digit.

Parse a Codice Fiscale

```
using CodiceFiscale;
using CodiceFiscale.Entities;
using CodiceFiscale.Enums;

CodiceFiscaleData data = CodiceFiscaleParser.Parse("RSSMRA85T10A562S");

Gender gender      = data.Gender;           // Gender.Male
DateOnly dob       = data.DateOfBirth;      // new DateOnly(1985, 12, 10)
string belfiore    = data.BelfioreCode;     // "A562"
```

⚠ WARNING

`Parse` throws `InvalidCodiceFiscaleException` if the input is not valid. Always validate first, or wrap in a try/catch.

```
try
{
    CodiceFiscaleData data = CodiceFiscaleParser.Parse(userInput);
}
catch (InvalidCodiceFiscaleException ex)
{
    Console.WriteLine(ex.Message);
}
```

Validate an IBAN

```
using CodiceFiscale;

bool valid = IBANValidator.IsValid("IT60X0542811101000000123456"); // true
bool validSpaced = IBANValidator.IsValid("IT60 X054 2811 1010 0000 0123 456"); // true
(spaces ignored)

bool invalid = IBANValidator.IsValid("IT00X0542811101000000123456"); // false (bad checksum)
```

Validate a Partita IVA

```
using CodiceFiscale;

// Standard company VAT
bool valid = ItalianVatCodeValidator.IsValid("00484960588", isConsumer: false, isFiscal:
false); // true

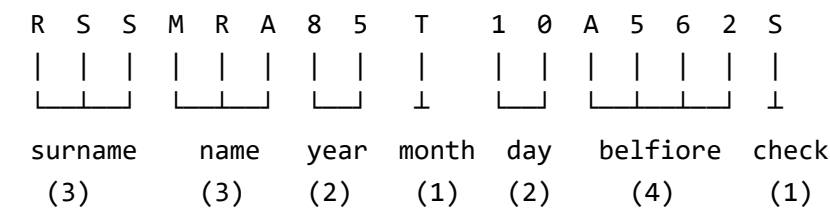
// Consumer (natural person) VAT – first digit must be 8 or 9
bool consumer = ItalianVatCodeValidator.IsValid("85423511618", isConsumer: true, isFiscal:
false); // true

// Accept a Codice Fiscale as a valid fiscal identifier
bool fiscal = ItalianVatCodeValidator.IsValid("RSSMRA85T10A562S", isConsumer: false,
isFiscal: true); // true
```


Codice Fiscale Structure

The Italian *Codice Fiscale* is a 16-character alphanumeric string that uniquely identifies a natural person for tax purposes.

Layout



Position	Length	Content	Example
1–3	3 chars	Surname consonants/vowels	RSS
4–6	3 chars	Name consonants/vowels	MRA
7–8	2 digits	Year of birth (last 2 digits)	85 → 1985
9	1 char	Month of birth (letter code)	T → December
10–11	2 digits	Day of birth (female: day + 40)	10 → day 10
12–15	4 chars	Belfiore catastal code	A562
16	1 char	Check character	S

Surname encoding (positions 1–3)

Consonants are taken first (in order), then vowels. If fewer than 3 characters are available, **x** is used as padding.

Surname	Consonants	Vowels	Code
Rossi	R, S, S	O, I	RSS
Re	R	E	REX
Fo	F	O	FOX

Name encoding (positions 4–6)

If the name has **4 or more consonants**, the 1st, 3rd, and 4th are taken. Otherwise, the same consonants-then-vowels-then-X rule applies as for the surname.

Name	Consonants	Code
Mario	M, R	MRA (M + R + vowel A)
Luca	L, C	LCU (L + C + vowel U)
Alessandro	L, S, S, N, D, R	LSN (1st, 3rd, 4th consonant)

Month encoding (position 9)

Letter	Month
A	January
B	February
C	March
D	April
E	May
H	June
L	July
M	August
P	September
R	October
S	November
T	December

Day encoding (positions 10–11)

- **Male:** actual day of birth (01–31)
- **Female:** day of birth + 40 (41–71)

This allows unambiguous gender determination from the two-digit day field.

Belfiore code (positions 12–15)

A 4-character code identifying the place of birth:

- **Italian municipality:** one letter (A–Z, excluding certain letters) + 3 digits, e.g. A562 (Rome)
- **Foreign country:** always starts with Z + 3 digits, e.g. Z614 (Venezuela)

The full dataset (~7 800 Italian municipalities + ~260 foreign countries) is embedded in the library.

Check character (position 16)

Computed from the first 15 characters using a weighted sum algorithm:

1. Characters at **odd positions** (1, 3, 5, ...) use a non-linear lookup table
2. Characters at **even positions** (2, 4, 6, ...) use their ordinal value (0–25 for letters, 0–9 for digits)
3. Sum all values, take modulo 26, convert to letter (A=0, B=1, ...)

This library validates the check character automatically inside `CodiceFiscaleValidator.IsValid`.

Omocodia

When two people would share the same Codice Fiscale (same name, birthdate, and birthplace), one or more of the numeric digits (positions 7–8, 10–11) are replaced with letters according to a fixed substitution table:

Digit	Letter
0	L
1	M
2	N
3	P
4	Q

Digit	Letter
5	R
6	S
7	T
8	U
9	V

** NOTE**

Omocodia support is not yet implemented in this library. Standard Codici Fiscali (all digits in numeric positions) are fully supported.

Validation

Overview

`CodiceFiscaleValidator.IsValid(string cf)` performs a multi-step validation of an Italian Codice Fiscale. It returns `true` only if **all** checks pass.

```
using CodiceFiscale;

bool valid = CodiceFiscaleValidator.IsValid("RSSMRA85T10A562S"); // true
```

The method is **static** and **thread-safe**. It never throws — invalid or `null`-like inputs always return `false`.

Validation pipeline

The validator applies the following checks in order, short-circuiting on the first failure:

Step 1 — Basic format

- Input must not be null or empty
- Length must be exactly **16 characters**
- Must contain **no whitespace** (tabs, spaces, newlines all fail)

```
CodiceFiscaleValidator.IsValid("RSSMRA85T10A562S "); // false (trailing space)
CodiceFiscaleValidator.IsValid("RSSMRA85T10A562"); // false (15 chars)
```

Step 2 — Regex structure

The input is matched against the pattern:

<code>[A-Z]{3}</code>	surname
<code>[A-Z]{3}</code>	name
<code>[0-9]{2}</code>	year
<code>[ABCDEHILMPRST]</code>	month letter
<code>[0-9]{2}</code>	day
<code>\S{4}</code>	belfiore code
<code>[A-Z]</code>	<code>check</code> character

Input is uppercased before matching, so lowercase letters are accepted.

```
CodiceFiscaleValidator.IsValid("rssmra85t10a562s"); // true
CodiceFiscaleValidator.IsValid("RSS1RA85T10A562S"); // false (digit in name)
```

```
CodiceFiscaleValidator.IsValid("RSSMRA85110A562S"); // false (digit in month)
```

Step 3 — Date coherence

The day and month are validated against the actual calendar:

- Months with 30 days (April **D**, June **H**, September **P**, November **S**): day 1–30 (male) or 41–70 (female)
- February **B**: day 1–28 or 1–29 for leap years (male) or 41–68/41–69 (female)
- Other months: day 1–31 (male) or 41–71 (female)

```
CodiceFiscaleValidator.IsValid("RSSMRA85D31A562?"); // false (April has 30 days)
CodiceFiscaleValidator.IsValid("RSSMRA85B29A562?"); // false (1985 is not a leap year)
```

Step 4 — Belfiore code

The 4-character Belfiore code is looked up in the embedded datasets:

- Codes starting with a **letter other than Z**: searched in the Italian municipalities dataset (~7 800 comuni)
- Codes starting with **Z**: searched in the foreign countries dataset (~260 countries)

```
CodiceFiscaleValidator.IsValid("RSSMRA85T10Z999S"); // false (Z999 not assigned)
CodiceFiscaleValidator.IsValid("MRNGRL01P55Z614X"); // true (Z614 = Venezuela)
```

Step 5 — Check character

The 16th character is recomputed from the first 15 using the standard algorithm and compared to the actual value. See [Codice Fiscale Structure](#) for details.

```
CodiceFiscaleValidator.IsValid("RSSMRA85T10A562S"); // true (S is correct)
CodiceFiscaleValidator.IsValid("RSSMRA85T10A562X"); // false (X is wrong)
```

Return value summary

Condition	Returns
All checks pass	true
Null, empty, or whitespace-only	false
Wrong length	false

Condition	Returns
Contains whitespace	false
Fails regex pattern	false
Impossible calendar date	false
Unknown Belfiore/Z code	false
Wrong check character	false

Parsing

Overview

`CodiceFiscaleParser.Parse(string cf)` decodes a valid Codice Fiscale into its constituent personal data.

```
using CodiceFiscale;  
using CodiceFiscale.Entities;  
using CodiceFiscale.Enums;
```

```
CodiceFiscaleData data = CodiceFiscaleParser.Parse("RSSMRA85T10A562S");
```

Return type: `CodiceFiscaleData`

```
public record CodiceFiscaleData(  
    Gender    Gender,  
    DateOnly  DateOfBirth,  
    string    BelfioreCode);
```

Property	Type	Description	Example
Gender	Gender	Male or Female	<code>Gender.Male</code>
DateOfBirth	DateOnly	Full date of birth	<code>new DateOnly(1985, 12, 10)</code>
BelfioreCode	string	4-char Belfiore code of place of birth	<code>"A562"</code>

Gender

Gender is inferred from the day field:

- Day **1–31** → `Gender.Male`
- Day **41–71** → `Gender.Female` (actual day = encoded day – 40)

Date of birth

The year is reconstructed from the 2-digit year using the current year as cutoff:

- If the 2-digit year ≤ current year's last 2 digits → 2000s (e.g. `01` → 2001)
- Otherwise → 1900s (e.g. `85` → 1985)

Belfiore code

The raw 4-character code is returned as-is. Italian municipality codes start with a letter (other than Z); foreign country codes start with **Z**.

Exception: **InvalidCodiceFiscaleException**

`Parse` runs `CodiceFiscaleValidator.IsValid` internally. If the input is not valid, it throws:

```
throw new InvalidCodiceFiscaleException($"The provided Codice Fiscale '{cf}' is not valid");
```

Always validate before parsing, or handle the exception:

```
// Option 1 – validate first
if (CodiceFiscaleValidator.IsValid(input))
{
    CodiceFiscaleData data = CodiceFiscaleParser.Parse(input);
}

// Option 2 – try/catch
try
{
    CodiceFiscaleData data = CodiceFiscaleParser.Parse(input);
}
catch (InvalidCodiceFiscaleException ex)
{
    // handle invalid input
}
```

Examples

```
// Male, born in Italy
CodiceFiscaleData mario = CodiceFiscaleParser.Parse("RSSMRA85T10A562S");
// Gender:      Male
// DateOfBirth:  1985-12-10
// BelfioreCode: A562 (Rome)

// Female, born abroad (Venezuela)
CodiceFiscaleData gabriella = CodiceFiscaleParser.Parse("MRNGRL01P55Z614X");
// Gender:      Female (55 - 40 = day 15)
// DateOfBirth:  2001-09-15
// BelfioreCode: Z614 (Venezuela)
```

IBAN Validation

Overview

`IBANValidator.IsValid(string iban)` validates an International Bank Account Number according to the ISO 13616 standard using the mod-97 algorithm.

```
using CodiceFiscale;
```

```
bool valid = IBANValidator.IsValid("IT60X0542811101000000123456"); // true
```

The method is **static** and **thread-safe**. It never throws.

Supported formats

Spaces are accepted and stripped before validation, so both compact and formatted IBANs are valid inputs:

```
IBANValidator.IsValid("IT60X0542811101000000123456");           // true – compact
IBANValidator.IsValid("IT60 X054 2811 1010 0000 0123 456");       // true – formatted
```

Validation algorithm

1. **Strip whitespace** from the input
 2. **Check country code** — the first 2 characters must be a valid ISO 3166-1 alpha-2 country code that participates in the IBAN scheme
 3. **Check expected length** — each country has a fixed IBAN length (e.g. Italy: 27, Germany: 22, UK: 22)
 4. **Rearrange** — move the first 4 characters to the end
 5. **Convert letters to digits** — A=10, B=11, ..., Z=35
 6. **Compute mod 97** — the result must equal **1**
-

Examples

```
// Valid IBANs
IBANValidator.IsValid("IT60X0542811101000000123456"); // Italy
IBANValidator.IsValid("DE89370400440532013000");      // Germany
IBANValidator.IsValid("GB29NWBK60161331926819");      // United Kingdom
IBANValidator.IsValid("FR7630006000011234567890189"); // France
IBANValidator.IsValid("ES9121000418450200051332");    // Spain
```

```
// Invalid IBANs
IBANValidator.IsValid(""); // false (empty)
IBANValidator.IsValid("IT00X0542811101000000123456"); // false (bad mod-97 checksum)
IBANValidator.IsValid("XX60X0542811101000000123456"); // false (unknown country XX)
IBANValidator.IsValid("IT60X054281110100000012345"); // false (wrong length for Italy)
```

VAT Validation (Partita IVA)

Overview

`ItalianVatCodeValidator.IsValid(string vat, bool isConsumer, bool isFiscal)` validates Italian VAT codes (*Partita IVA*) according to the rules defined by the Agenzia delle Entrate.

```
using CodiceFiscale;
```

```
bool valid = ItalianVatCodeValidator.IsValid("00484960588", isConsumer: false,
isFiscal: false);
// true
```

The method is **static** and **thread-safe**. It never throws.

Parameters

Parameter	Type	Description
<code>vat</code>	<code>string</code>	The code to validate. Either an 11-digit Partita IVA or a 16-character Codice Fiscale (if <code>isFiscal</code> is <code>true</code>).
<code>isConsumer</code>	<code>bool</code>	<code>true</code> if the code belongs to a natural person (consumer). The first digit must be 8 or 9.
<code>isFiscal</code>	<code>bool</code>	<code>true</code> to also accept a 16-character Codice Fiscale as a valid identifier.

Validation algorithm

A standard 11-digit Partita IVA is validated as follows:

1. **Length** must be exactly 11 digits (all numeric)
2. **Consumer flag** — if `isConsumer: true`, first digit must be 8 or 9; if `false`, first digit must be 0–7
3. **Luhn-like checksum** (Agenzia delle Entrate algorithm):
 - Sum digits at **odd positions** (1, 3, 5, 7, 9) directly
 - Double digits at **even positions** (2, 4, 6, 8, 10); if the result ≥ 10 , subtract 9
 - The 11th digit must equal $(10 - (\text{sum} \% 10)) \% 10$

Mode combinations

isConsumer	isFiscal	Accepts
false	false	11-digit VAT, first digit 0–7
true	false	11-digit VAT, first digit 8–9
false	true	11-digit VAT (any) or valid 16-char Codice Fiscale
true	true	Consumer VAT (first digit 8–9) or valid Codice Fiscale

Examples

```
// Standard company VAT
```

```
ItalianVatCodeValidator.IsValid("00484960588", false, false); // true
```

```
ItalianVatCodeValidator.IsValid("10433218194", false, false); // true
```

```
// Consumer VAT (first digit 8 or 9)
```

```
ItalianVatCodeValidator.IsValid("85423511618", true, false); // true
```

```
ItalianVatCodeValidator.IsValid("91849593107", true, false); // true
```

```
// Codice Fiscale accepted as fiscal identifier
```

```
ItalianVatCodeValidator.IsValid("RSSMRA85T10A562S", false, true); // true
```

```
ItalianVatCodeValidator.IsValid("00484960588", false, true); // true (VAT  
also accepted)
```

```
// Invalid cases
```

```
ItalianVatCodeValidator.IsValid("", false, false); // false (empty)
```

```
ItalianVatCodeValidator.IsValid(" ", false, false); // false (whitespace)
```

```
ItalianVatCodeValidator.IsValid("0218208039", false, false); // false (10 digits,  
too short)
```

```
ItalianVatCodeValidator.IsValid("021820803900", false, false); // false (12 digits, too long)
```

```
ItalianVatCodeValidator.IsValid("02182080391", false, false); // false (bad checksum)
```

```
ItalianVatCodeValidator.IsValid("80025291002", false, false); // false (first digit 8,  
but isConsumer=false)
```

```
ItalianVatCodeValidator.IsValid("02182080390", true, false); // false (first digit 0,  
but isConsumer=true)
```

```
ItalianVatCodeValidator.IsValid("RSSMRA85T10A562S", false, false); // false (CF not accepted  
without isFiscal)
```